

Chapter 2: State Values and Bellman Equation

Adapted from Mathematical Foundations of RL by Shiyu Zhao

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Chapter Overview

Core Concept: State Value

The **state value** is defined as the average reward that an agent can obtain if it follows a given policy.

- The greater the state value, the better the corresponding policy
- State values serve as a metric to evaluate policies

Important Tool: Bellman Equation

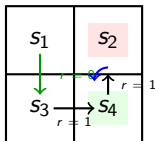
The **Bellman equation** describes the relationships between the values of all states.

- By solving the Bellman equation, we obtain state values
- This process is called **policy evaluation**

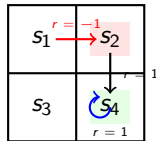
Additional Concept

The **action value** indicates the “value” of taking an action at a state.

Three Policies at State s_1



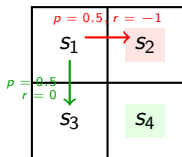
Policy 1: Go down from s_1



Policy 2: Go right from s_1

- **Policy 1:** Avoids forbidden area $\Rightarrow$ Intuitively **best**
- **Policy 2:** Goes to forbidden area $\Rightarrow$ Intuitively **worse**

Stochastic Policy



Policy 3: 50% right, 50% down

Intuition:

- Policy 3 is “in between” Policies 1 and 2
- Has 50% probability to go to forbidden area

Question: Can we use mathematics to describe this intuition?

Answer: Yes! Using [returns](#).

Calculating Returns for Each Policy

Return for Policy 1: Trajectory $s_1 \rightarrow s_3 \rightarrow s_4 \rightarrow s_4 \cdots$

$$\begin{aligned}\text{return}_1 &= 0 + \gamma \cdot 1 + \gamma^2 \cdot 1 + \dots \\ &= \gamma(1 + \gamma + \gamma^2 + \dots) \\ &= \boxed{\frac{\gamma}{1 - \gamma}}\end{aligned}$$

Return for Policy 2: Trajectory $s_1 \rightarrow s_2 \rightarrow s_4 \rightarrow s_4 \cdots$

$$\begin{aligned}\text{return}_2 &= -1 + \gamma \cdot 1 + \gamma^2 \cdot 1 + \dots \\ &= -1 + \gamma(1 + \gamma + \gamma^2 + \dots) \\ &= \boxed{-1 + \frac{\gamma}{1 - \gamma}}\end{aligned}$$

where $\gamma \in (0, 1)$ is the discount rate.

Return for Stochastic Policy

Return for Policy 3: Two possible trajectories with probability 0.5 each

- Trajectory 1: $s_1 \rightarrow s_3 \rightarrow s_4 \rightarrow s_4 \cdots$ (with prob. 0.5)
- Trajectory 2: $s_1 \rightarrow s_2 \rightarrow s_4 \rightarrow s_4 \cdots$ (with prob. 0.5)

The **average return** starting from s_1 :

$$\begin{aligned} \text{return}_3 &= 0.5 \left(-1 + \frac{\gamma}{1-\gamma} \right) + 0.5 \left(\frac{\gamma}{1-\gamma} \right) \\ &= \boxed{-0.5 + \frac{\gamma}{1-\gamma}} \end{aligned}$$

Key Comparison

For any value of $\gamma \in (0, 1)$:

$$\text{return}_1 > \text{return}_3 > \text{return}_2$$

Note: return_3 is actually a **state value** (expected return).

Conclusion: Returns Evaluate Policies

$$\text{return}_1 > \text{return}_3 > \text{return}_2$$

- Policy 1 is **best** (greatest return)
- Policy 2 is **worst** (smallest return)

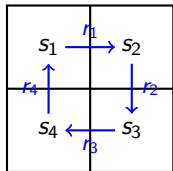
Consistency with Intuition

- Policy 1 avoids the forbidden area $\Rightarrow$ best
- Policy 2 leads to the forbidden area $\Rightarrow$ worst
- Policy 3 is in between $\Rightarrow$ intermediate

Key Insight

Returns can be used to **evaluate policies**: A policy is better if the return obtained by following that policy is greater.

Two Ways to Calculate Returns



Cyclic trajectory

Method 1: By Definition

Returns as discounted sum of rewards:

$$v_1 = r_1 + \gamma r_2 + \gamma^2 r_3 + \dots$$

$$v_2 = r_2 + \gamma r_3 + \gamma^2 r_4 + \dots$$

$$v_3 = r_3 + \gamma r_4 + \gamma^2 r_1 + \dots$$

$$v_4 = r_4 + \gamma r_1 + \gamma^2 r_2 + \dots$$

where v_i = return starting from s_i

Method 2: Bootstrapping

Rewriting the returns:

$$v_1 = r_1 + \gamma(r_2 + \gamma r_3 + \dots) = r_1 + \gamma v_2$$

$$v_2 = r_2 + \gamma(r_3 + \gamma r_4 + \dots) = r_2 + \gamma v_3$$

$$v_3 = r_3 + \gamma(r_4 + \gamma r_1 + \dots) = r_3 + \gamma v_4$$

$$v_4 = r_4 + \gamma(r_1 + \gamma r_2 + \dots) = r_4 + \gamma v_1$$

Bootstrapping

The values **rely on each other**:

- v_1 relies on v_2
- v_2 relies on v_3
- v_3 relies on v_4
- v_4 relies on v_1

Bootstrapping = obtaining values of quantities from themselves

Matrix-Vector Form of Bootstrapping

The bootstrapping equations can be written in matrix-vector form:

$$\underbrace{\begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix}}_v = \underbrace{\begin{bmatrix} r_1 \\ r_2 \\ r_3 \\ r_4 \end{bmatrix}}_r + \underbrace{\begin{bmatrix} \gamma v_2 \\ \gamma v_3 \\ \gamma v_4 \\ \gamma v_1 \end{bmatrix}}_{\gamma P v}$$

Which expands to:

$$\begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix} = \begin{bmatrix} r_1 \\ r_2 \\ r_3 \\ r_4 \end{bmatrix} + \underbrace{\gamma \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix}}_P \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix}$$

Compact Form

$$v = r + \gamma P v \quad \Rightarrow \quad v = (I - \gamma P)^{-1} r$$

Key Insight: The Bellman Equation

Core Idea

The return obtained by starting from one state **depends on** those obtained when starting from other states.

This is the essence of the **Bellman equation**!

- The matrix P represents state transitions
- The equation $v = r + \gamma P v$ describes how values are related
- Can be solved: $v = (I - \gamma P)^{-1} r$

Question: Is $I - \gamma P$ always invertible?

Answer: Yes! (Explained in Section 2.7.1 in the book)

Why State Values?

Limitation of Returns

Returns are inapplicable to **stochastic systems** because starting from one state may lead to **different returns**.

Solution: Use **state values** (expected returns)

Notation: At time t :

- S_t : current state (random variable)
- A_t : action taken following policy π
- S_{t+1} : next state
- R_{t+1} : immediate reward

Process: $S_t \xrightarrow{A_t} S_{t+1}, R_{t+1}$

Note: $S_t, S_{t+1} \in \mathcal{S}$, $A_t \in \mathcal{A}(S_t)$, $R_{t+1} \in \mathcal{R}(S_t, A_t)$

Definition of State Value

Trajectory starting from time t :

$$S_t \xrightarrow{A_t} S_{t+1}, R_{t+1} \xrightarrow{A_{t+1}} S_{t+2}, R_{t+2} \xrightarrow{A_{t+2}} S_{t+3}, R_{t+3} \dots$$

Discounted Return:

$$G_t \triangleq R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots$$

where $\gamma \in (0, 1)$ is the discount rate.

State Value Definition

$$v_{\pi}(s) \triangleq \mathbb{E}[G_t | S_t = s]$$

This is called the **state-value function** or simply the **state value** of s .

Note: G_t is a random variable since $R_{t+1}, R_{t+2}, \dots$ are all random variables.

Properties of State Values

① $v_{\pi}(s)$ **depends on** s

- Definition is a conditional expectation with condition $S_t = s$

② $v_{\pi}(s)$ **depends on** π

- Trajectories are generated by following policy π
- Different policy $\Rightarrow$ different state value

③ $v_{\pi}(s)$ **does NOT depend on** t

- The value is determined once the policy is given
- t represents the current time step

State Values vs Returns

Deterministic Case:

- Both policy and system are deterministic
- Starting from a state $\Rightarrow$ same trajectory
- $\text{Return} = \text{State Value}$

Stochastic Case:

- Either policy or system is stochastic
- Same starting state $\Rightarrow$ different trajectories
- Different trajectories $\Rightarrow$ different returns
- $\text{State Value} = \text{Mean of Returns}$

Policy Evaluation

State values constitute a **core concept** in RL:

- Policies that generate **greater state values** are **better**
- More formal than using returns directly

Deriving the Bellman Equation

Step 1: Rewrite G_t recursively

$$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots = R_{t+1} + \gamma G_{t+1}$$

where $G_{t+1} = R_{t+2} + \gamma R_{t+3} + \dots$

Step 2: Apply expectation

$$\begin{aligned} v_{\pi}(s) &= \mathbb{E}[G_t | S_t = s] \\ &= \mathbb{E}[R_{t+1} + \gamma G_{t+1} | S_t = s] \\ &= \underbrace{\mathbb{E}[R_{t+1} | S_t = s]}_{\text{immediate rewards}} + \gamma \underbrace{\mathbb{E}[G_{t+1} | S_t = s]}_{\text{future rewards}} \end{aligned}$$

Now we analyze each term separately...

Term 1: Expected Immediate Rewards

Using the **law of total expectation**:

$$\begin{aligned}\mathbb{E}[R_{t+1}|S_t = s] &= \sum_{a \in \mathcal{A}} \pi(a|s) \mathbb{E}[R_{t+1}|S_t = s, A_t = a] \\ &= \sum_{a \in \mathcal{A}} \pi(a|s) \sum_{r \in \mathcal{R}} p(r|s, a) \cdot r\end{aligned}$$

Explanation:

- $\pi(a|s)$: probability of taking action a in state s under policy π
- $p(r|s, a)$: probability of receiving reward r given state s and action a
- $\mathcal{A}$: set of possible actions
- $\mathcal{R}$: set of possible rewards

Term 2: Expected Future Rewards

$$\begin{aligned}\mathbb{E}[G_{t+1}|S_t = s] &= \sum_{s' \in \mathcal{S}} \mathbb{E}[G_{t+1}|S_t = s, S_{t+1} = s'] \cdot p(s'|s) \\ &= \sum_{s' \in \mathcal{S}} \mathbb{E}[G_{t+1}|S_{t+1} = s'] \cdot p(s'|s) \quad (\text{Markov property}) \\ &= \sum_{s' \in \mathcal{S}} v_{\pi}(s') \cdot p(s'|s) \\ &= \sum_{s' \in \mathcal{S}} v_{\pi}(s') \sum_{a \in \mathcal{A}} p(s'|s, a) \pi(a|s)\end{aligned}$$

Markov Property

Future rewards depend only on the **present state**, not on previous states:

$$\mathbb{E}[G_{t+1}|S_t = s, S_{t+1} = s'] = \mathbb{E}[G_{t+1}|S_{t+1} = s']$$

The Bellman Equation

Bellman Equation

Combining the two terms:

$$\begin{aligned} v_{\pi}(s) = & \underbrace{\sum_{a \in \mathcal{A}} \pi(a|s) \sum_{r \in \mathcal{R}} p(r|s, a) r}_{\text{mean of immediate rewards}} \\ & + \underbrace{\gamma \sum_{a \in \mathcal{A}} \pi(a|s) \sum_{s' \in \mathcal{S}} p(s'|s, a) v_{\pi}(s')}_{\text{mean of future rewards}} \end{aligned}$$

Or equivalently:

$$v_{\pi}(s) = \sum_{a \in \mathcal{A}} \pi(a|s) \left[\sum_{r \in \mathcal{R}} p(r|s, a) r + \gamma \sum_{s' \in \mathcal{S}} p(s'|s, a) v_{\pi}(s') \right]$$

for all $s \in \mathcal{S}$

Understanding the Bellman Equation

Key components:

- $v_{\pi}(s)$ and $v_{\pi}(s')$: **unknown** state values to be calculated
- $\pi(a|s)$: **given** policy
- $p(r|s, a)$ and $p(s'|s, a)$: **system model**

Important Notes

- 1 The Bellman equation is a **set of linear equations** for all states, not a single equation
- 2 Solving for v_{π} from the Bellman equation is called **policy evaluation**
- 3 We will see how to solve without the model using **model-free algorithms**

Alternative Forms of the Bellman Equation

Form 1: Using joint probability $p(s', r|s, a)$

From: $p(s'|s, a) = \sum_{r \in \mathcal{R}} p(s', r|s, a)$ and $p(r|s, a) = \sum_{s' \in \mathcal{S}} p(s', r|s, a)$

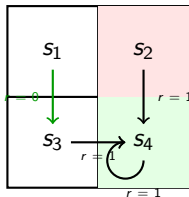
$$v_{\pi}(s) = \sum_{a \in \mathcal{A}} \pi(a|s) \sum_{s' \in \mathcal{S}} \sum_{r \in \mathcal{R}} p(s', r|s, a) [r + \gamma v_{\pi}(s')]$$

Form 2: When reward depends only on next state s'

If $r = r(s')$, then $p(r(s')|s, a) = p(s'|s, a)$:

$$v_{\pi}(s) = \sum_{a \in \mathcal{A}} \pi(a|s) \sum_{s' \in \mathcal{S}} p(s'|s, a) [r(s') + \gamma v_{\pi}(s')]$$

Example 1: Deterministic Policy



For state s_1 :

- $\pi(a = a_3 | s_1) = 1, \pi(a \neq a_3 | s_1) = 0$
- $p(s' = s_3 | s_1, a_3) = 1$
- $p(r = 0 | s_1, a_3) = 1$

Substituting into Bellman equation:

$$v_{\pi}(s_1) = 0 + \gamma v_{\pi}(s_3)$$

Example 1: Complete System of Equations

Bellman equations for all states:

$$v_{\pi}(s_1) = 0 + \gamma v_{\pi}(s_3)$$

$$v_{\pi}(s_2) = 1 + \gamma v_{\pi}(s_4)$$

$$v_{\pi}(s_3) = 1 + \gamma v_{\pi}(s_4)$$

$$v_{\pi}(s_4) = 1 + \gamma v_{\pi}(s_4)$$

Solving the equations:

$$v_{\pi}(s_4) = \frac{1}{1 - \gamma}$$

$$v_{\pi}(s_3) = \frac{1}{1 - \gamma}$$

$$v_{\pi}(s_2) = \frac{1}{1 - \gamma}$$

$$v_{\pi}(s_1) = \frac{\gamma}{1 - \gamma}$$

Example 1: Numerical Results

Setting $\gamma = 0.9$:

$$v_{\pi}(s_4) = \frac{1}{1 - 0.9} = 10$$

$$v_{\pi}(s_3) = \frac{1}{1 - 0.9} = 10$$

$$v_{\pi}(s_2) = \frac{1}{1 - 0.9} = 10$$

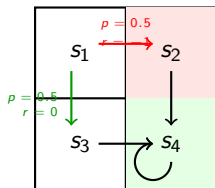
$$v_{\pi}(s_1) = \frac{0.9}{1 - 0.9} = 9$$

Observation

$$v_{\pi}(s_1) = 9 < v_{\pi}(s_2) = v_{\pi}(s_3) = v_{\pi}(s_4) = 10$$

This is because s_1 gets reward 0 on its first transition, while others get reward 1.

Example 2: Stochastic Policy



For state s_1 :

- $\pi(a_2|s_1) = 0.5$ (go right)
- $\pi(a_3|s_1) = 0.5$ (go down)
- $p(s_3|s_1, a_3) = 1$, $p(s_2|s_1, a_2) = 1$
- $p(r = 0|s_1, a_3) = 1$, $p(r = -1|s_1, a_2) = 1$

Bellman equation:

$$v_{\pi}(s_1) = 0.5[0 + \gamma v_{\pi}(s_3)] + 0.5[-1 + \gamma v_{\pi}(s_2)]$$

Example 2: Solution

System of equations:

$$v_{\pi}(s_1) = 0.5[0 + \gamma v_{\pi}(s_3)] + 0.5[-1 + \gamma v_{\pi}(s_2)]$$

$$v_{\pi}(s_2) = 1 + \gamma v_{\pi}(s_4)$$

$$v_{\pi}(s_3) = 1 + \gamma v_{\pi}(s_4)$$

$$v_{\pi}(s_4) = 1 + \gamma v_{\pi}(s_4)$$

Solutions:

$$v_{\pi}(s_4) = \frac{1}{1 - \gamma}, \quad v_{\pi}(s_3) = \frac{1}{1 - \gamma}, \quad v_{\pi}(s_2) = \frac{1}{1 - \gamma}$$

$$v_{\pi}(s_1) = -0.5 + \frac{\gamma}{1 - \gamma}$$

With $\gamma = 0.9$:

$$v_{\pi}(s_4) = v_{\pi}(s_3) = v_{\pi}(s_2) = 10, \quad v_{\pi}(s_1) = -0.5 + 9 = 8.5$$

Comparing the Two Policies

Policy Comparison

For all states $i = 1, 2, 3, 4$:

$$v_{\pi_1}(s_i) \geq v_{\pi_2}(s_i)$$

State	Policy 1 (Deterministic)	Policy 2 (Stochastic)
s_1	9	8.5
s_2	10	10
s_3	10	10
s_4	10	10

Conclusion

Policy 1 is **better** because it has greater state values.

This is consistent with intuition: Policy 1 avoids the forbidden area from s_1 .

Rewriting the Bellman Equation

Define:

$$r_{\pi}(s) \triangleq \sum_{a \in \mathcal{A}} \pi(a|s) \sum_{r \in \mathcal{R}} p(r|s, a) r \quad (\text{mean immediate reward})$$

$$p_{\pi}(s'|s) \triangleq \sum_{a \in \mathcal{A}} \pi(a|s) p(s'|s, a) \quad (\text{transition probability under } \pi)$$

Simplified Bellman equation:

$$v_{\pi}(s) = r_{\pi}(s) + \gamma \sum_{s' \in \mathcal{S}} p_{\pi}(s'|s) v_{\pi}(s')$$

This is valid for each state s_i :

$$v_{\pi}(s_i) = r_{\pi}(s_i) + \gamma \sum_{s_j \in \mathcal{S}} p_{\pi}(s_j|s_i) v_{\pi}(s_j)$$

Matrix-Vector Notation

Definitions: Let $n = |\mathcal{S}|$ (number of states)

- $v_\pi = [v_\pi(s_1), \dots, v_\pi(s_n)]^T \in \mathbb{R}^n$ (state value vector)
- $r_\pi = [r_\pi(s_1), \dots, r_\pi(s_n)]^T \in \mathbb{R}^n$ (reward vector)
- $P_\pi \in \mathbb{R}^{n \times n}$ with $[P_\pi]_{ij} = p_\pi(s_j | s_i)$ (transition matrix)

Matrix-Vector Form of Bellman Equation

$$v_\pi = r_\pi + \gamma P_\pi v_\pi$$

where:

- v_π is **unknown** to be solved
- r_π, P_π are **known**

Properties of P_π

Property 1: Nonnegative Matrix

$$P_\pi \geq 0 \text{ (all elements } \geq 0)$$

This is because transition probabilities are always nonnegative.

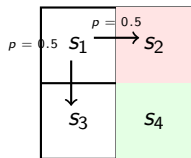
Property 2: Stochastic Matrix

$$P_\pi \mathbf{1} = \mathbf{1}, \text{ where } \mathbf{1} = [1, \dots, 1]^T$$

The sum of each row equals 1 (probability distribution over next states).

These properties ensure that the Bellman equation has well-defined solutions.

Example: Matrix-Vector Form



$$\begin{bmatrix} v_{\pi}(s_1) \\ v_{\pi}(s_2) \\ v_{\pi}(s_3) \\ v_{\pi}(s_4) \end{bmatrix} = \begin{bmatrix} r_{\pi}(s_1) \\ r_{\pi}(s_2) \\ r_{\pi}(s_3) \\ r_{\pi}(s_4) \end{bmatrix} + \gamma P_{\pi} \begin{bmatrix} v_{\pi}(s_1) \\ v_{\pi}(s_2) \\ v_{\pi}(s_3) \\ v_{\pi}(s_4) \end{bmatrix}$$

$$\begin{bmatrix} v_{\pi}(s_1) \\ v_{\pi}(s_2) \\ v_{\pi}(s_3) \\ v_{\pi}(s_4) \end{bmatrix} = \begin{bmatrix} 0.5(0) + 0.5(-1) \\ 1 \\ 1 \\ 1 \end{bmatrix} + \gamma \begin{bmatrix} 0 & 0.5 & 0.5 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} v_{\pi}(s_1) \\ v_{\pi}(s_2) \\ v_{\pi}(s_3) \\ v_{\pi}(s_4) \end{bmatrix}$$

Verify: $P_{\pi} \mathbf{1} = \mathbf{1} \checkmark$

Method 1: Closed-Form Solution

From $v_\pi = r_\pi + \gamma P_\pi v_\pi$:

$$v_\pi - \gamma P_\pi v_\pi = r_\pi$$

$$(I - \gamma P_\pi) v_\pi = r_\pi$$

$$v_\pi = (I - \gamma P_\pi)^{-1} r_\pi$$

Closed-Form Solution

$$v_\pi = (I - \gamma P_\pi)^{-1} r_\pi$$

Properties of $(I - \gamma P_\pi)^{-1}$

Property 1: $I - \gamma P_\pi$ is invertible

Proof: Using the **Gershgorin circle theorem**:

- Every eigenvalue lies within at least one Gershgorin circle
- The i -th circle has center $[I - \gamma P_\pi]_{ii} = 1 - \gamma p_\pi(s_i|s_i)$
- Radius: $\sum_{j \neq i} |[I - \gamma P_\pi]_{ij}| = \sum_{j \neq i} \gamma p_\pi(s_j|s_i)$
- Since $\gamma < 1$: radius < center magnitude
- No circle contains the origin $\Rightarrow$ no zero eigenvalue

Property 2: $(I - \gamma P_\pi)^{-1} \geq I$

Since $P_\pi \geq 0$:

$$(I - \gamma P_\pi)^{-1} = I + \gamma P_\pi + \gamma^2 P_\pi^2 + \dots \geq I \geq 0$$

Property 3: Monotonicity

Property 3

For any vector $r \geq 0$:

$$(I - \gamma P_\pi)^{-1} r \geq r \geq 0$$

Proof:

- From Property 2: $(I - \gamma P_\pi)^{-1} - I \geq 0$
- Therefore: $[(I - \gamma P_\pi)^{-1} - I]r \geq 0$
- Which gives: $(I - \gamma P_\pi)^{-1} r \geq r$

Consequence

If $r_1 \geq r_2$, then:

$$(I - \gamma P_\pi)^{-1} r_1 \geq (I - \gamma P_\pi)^{-1} r_2$$

Method 2: Iterative Solution

Problem with closed-form: Matrix inversion is computationally expensive.

Iterative Algorithm

$$v_{k+1} = r_\pi + \gamma P_\pi v_k, \quad k = 0, 1, 2, \dots$$

Starting from initial guess $v_0 \in \mathbb{R}^n$:

$$v_k \rightarrow v_\pi = (I - \gamma P_\pi)^{-1} r_\pi \quad \text{as } k \rightarrow \infty$$

Advantages:

- No matrix inversion required
- Each iteration is simple: matrix-vector multiplication
- Converges for any initial guess v_0

Convergence Proof

Define error: $\delta_k = v_k - v_\pi$

Goal: Show $\delta_k \rightarrow 0$

Proof:

$$v_{k+1} = r_\pi + \gamma P_\pi v_k$$

$$\delta_{k+1} + v_\pi = r_\pi + \gamma P_\pi (\delta_k + v_\pi)$$

$$\delta_{k+1} = -v_\pi + r_\pi + \gamma P_\pi \delta_k + \gamma P_\pi v_\pi$$

$$\delta_{k+1} = \gamma P_\pi \delta_k \quad (\text{since } v_\pi = r_\pi + \gamma P_\pi v_\pi)$$

By induction:

$$\delta_{k+1} = \gamma^{k+1} P_\pi^{k+1} \delta_0$$

Since $0 \leq P_\pi^k \leq 1$ (elementwise) and $\gamma < 1$:

$$\gamma^{k+1} \rightarrow 0 \Rightarrow \delta_{k+1} \rightarrow 0$$

Illustrative Examples: Good vs Bad Policies

“Good” Policies:

3.5	3.9	4.3	4.8	5.3
3.1	3.5	4.8	5.3	5.9
2.8	2.5	10.0	5.9	6.6
2.5	10.0	10.0	10.0	7.3
2.3	9.0	10.0	9.0	8.1

Positive state values, reaching target

Settings: $r_{\text{boundary}} = r_{\text{forbidden}} = -1$, $r_{\text{target}} = 1$, $\gamma = 0.9$

“Bad” Policies:

-6.6	-7.3	-8.1	-9.0	-10
-8.5	-8.3	-8.1	-9.0	-10
-7.5	-8.3	-8.1	-9.0	-10
-7.5	-7.2	-9.1	-9.0	-10
-7.6	-7.3	-8.1	-9.0	-10

Negative state values, going to forbidden areas

Definition of Action Value

Action Value Definition

$$q_{\pi}(s, a) \triangleq \mathbb{E}[G_t | S_t = s, A_t = a]$$

The **action value** of a state-action pair (s, a) is the expected return after taking action a in state s .

Important notes:

- $q_{\pi}(s, a)$ depends on a **state-action pair**, not just the action
- More rigorously called “state-action value”
- Conventionally called “action value” for simplicity

Why study action values?

- They play a **more direct role** when finding optimal policies
- Heavily relies on state values (best to understand state values first)

Relationship: Action Values $\leftrightarrow$ State Values

Direction 1: State value from action values

Equation (2.13)

$$v_{\pi}(s) = \sum_{a \in \mathcal{A}} \pi(a|s) q_{\pi}(s, a)$$

A state value is the **expectation of action values** for that state.

Derivation:

$$\mathbb{E}[G_t | S_t = s] = \sum_{a \in \mathcal{A}} \mathbb{E}[G_t | S_t = s, A_t = a] \cdot \pi(a|s)$$

Relationship: State Values $\rightarrow$ Action Values

Direction 2: Action value from state values

Equation (2.14)

$$q_{\pi}(s, a) = \sum_{r \in \mathcal{R}} p(r|s, a)r + \gamma \sum_{s' \in \mathcal{S}} p(s'|s, a)v_{\pi}(s')$$

Components:

- First term: **mean of immediate rewards**
- Second term: **mean of future rewards**

Two sides of the same coin:

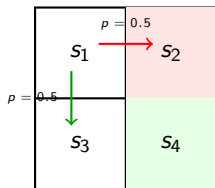
- (2.13): State values **from** action values
- (2.14): Action values **from** state values

Example: Calculating Action Values

Actions taken by policy at s_1 :

$$q_{\pi}(s_1, a_2) = -1 + \gamma v_{\pi}(s_2) \quad (\text{go right})$$

$$q_{\pi}(s_1, a_3) = 0 + \gamma v_{\pi}(s_3) \quad (\text{go down})$$



Actions NOT taken by policy at s_1 :

$$q_{\pi}(s_1, a_1) = -1 + \gamma v_{\pi}(s_1) \quad (\text{go up, bounce})$$

$$q_{\pi}(s_1, a_4) = -1 + \gamma v_{\pi}(s_1) \quad (\text{go left, bounce})$$

$$q_{\pi}(s_1, a_5) = 0 + \gamma v_{\pi}(s_1) \quad (\text{stay})$$

Common Mistake: Actions Not Selected by Policy

Wrong Reasoning

“Since policy π does not select a_1, a_4, a_5 at s_1 , we don't need their action values, or we can set $q_\pi(s_1, a_1) = q_\pi(s_1, a_4) = q_\pi(s_1, a_5) = 0$.”

Why this is wrong:

- ❶ **Every action has an action value**, regardless of whether the policy selects it
 - We can always calculate what would happen if we took that action
- ❷ **We need to explore all actions to find better policies**
 - The given policy might not be optimal
 - Some unselected actions might actually be better
 - RL aims to find optimal policies through exploration

Verifying State Value from Action Values

Using equation (2.13):

$$\begin{aligned}v_{\pi}(s_1) &= \sum_{a \in \mathcal{A}} \pi(a|s_1) q_{\pi}(s_1, a) \\&= 0.5 \cdot q_{\pi}(s_1, a_2) + 0.5 \cdot q_{\pi}(s_1, a_3) \\&= 0.5[-1 + \gamma v_{\pi}(s_2)] + 0.5[0 + \gamma v_{\pi}(s_3)]\end{aligned}$$

This matches our earlier derivation from the Bellman equation!

Consistency Check

The relationship between state values and action values is consistent with the Bellman equation.

Bellman Equation in Terms of Action Values

Substituting (2.13) into (2.14):

$$q_{\pi}(s, a) = \sum_{r \in \mathcal{R}} p(r|s, a)r + \gamma \sum_{s' \in \mathcal{S}} p(s'|s, a) \sum_{a' \in \mathcal{A}(s')} \pi(a'|s') q_{\pi}(s', a')$$

Matrix-Vector Form

$$q_{\pi} = \tilde{r} + \gamma P \Pi q_{\pi}$$

where:

- q_{π} : action value vector indexed by (s, a) pairs
- $\tilde{r}$: immediate reward vector, $[\tilde{r}]_{(s,a)} = \sum_r p(r|s, a)r$
- P : transition matrix, $[P]_{(s,a),s'} = p(s'|s, a)$
- Π : block diagonal matrix, $\Pi_{s',(s',a')} = \pi(a'|s')$

Key Features of Action Value Bellman Equation

Unique Properties

- $\tilde{r}$ and P are **independent of the policy**
 - Determined solely by the system model
- The policy π is embedded in Π
- The equation is also a **contraction mapping**
- Has a **unique solution** that can be iteratively solved

Comparison

	State Value Form	Action Value Form
Equation	$v_\pi = r_\pi + \gamma P_\pi v_\pi$	$q_\pi = \tilde{r} + \gamma P \Pi q_\pi$
Policy in	r_π and P_π	Π only
Model in	r_π and P_π	$\tilde{r}$ and P

Summary: Key Concepts

State Value

- Expected return starting from a state
- Used to evaluate policies (greater = better)
- Foundation for optimal policy definition

Bellman Equation

- Describes relationships among all state values
- Set of linear equations: $v_{\pi} = r_{\pi} + \gamma P_{\pi} v_{\pi}$
- Solving it = **policy evaluation**

Action Value

- Expected return after taking an action at a state
- More direct role in finding optimal policies
- Related to state values via two complementary equations

Summary: Key Methods

Solving the Bellman Equation

Closed-form: $v_\pi = (I - \gamma P_\pi)^{-1} r_\pi$

- Good for theoretical analysis
- Expensive matrix inversion

Iterative: $v_{k+1} = r_\pi + \gamma P_\pi v_k$

- Practical for computation
- Converges for any initial guess
- Error: $\delta_k = \gamma^k P_\pi^k \delta_0 \rightarrow 0$

Bootstrapping

Values depend on each other in a recursive manner. The matrix-vector form makes this explicit and solvable.

Next Chapter: Bellman Optimality Equation

- How to find the **optimal policy**?
- What makes a policy optimal?
- Optimal state values and action values

Key connections:

- Policy evaluation (this chapter) $\rightarrow$ Policy improvement
- Bellman equation $\rightarrow$ Bellman optimality equation
- State/action values $\rightarrow$ Optimal state/action values

Thank You!

Questions?